

PAH-OMC-015: counting-ensemble cutoff nondegeneracy

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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Purpose and scope

This checkpoint tests one additional researcher hypothesis: counting reference measure on the already declared full-charge PAH domain. The functional, rates, mobility and anchored strip are unchanged. The source dictionary and all nine parent hashes are in `PAH-OMC-015-counting-prereg-v1.json`, with SHA-256

0e07bd05c56c9765f15074505a5ce7916222
82a0e0c884bba277e780bbda0b35

The line break in this displayed hash is typographic only. The full derivation and source crosswalk are in the associated `PAH-OMC-015-cutoff-certificate.md`.

Let G_n be the OMC-004 strip, $n \geq 2$, with $N = 2(n + 2)$ vertices. Use $K = 2$, $M_s = M_\psi = 1$, $\epsilon = 1/2$, $\beta = \nu = 1$, $m^2 = 0$, and every other displayed coupling equal to one. First let the positive integer $R = R_{\max}$ tend to infinity at each fixed n , then let n tend to infinity. This restricted OMC-010 path does not remove all local cutoffs jointly and is not a physical continuum limit.

Every vertex has aperture, occupation and phase bits (j_v, ℓ_v, k_v) ; every representative edge has one link bit u_e . Phase labels at $\ell_v = 0$ remain distinct. Set $s_v = (1 + j_v)/2$, $\psi_v = R\ell_v(-1)^{k_v}$ and $U_e = (-1)^{u_e}$. The deterministic grade $Q = \sum_v \ell_v$ adds no multiplicity. There is no tagged probe in this task and no quotient of the counting space.

Finite ensemble and stationarity

For each nonempty grade define

$$Z_Q = \sum_{q(x)=Q} e^{-F_R(x)}, \quad Z = \sum_Q Z_Q, \quad w_Q = \frac{Z_Q}{Z}.$$

All sums are finite and strictly positive. The mixture of the original component Gibbs laws satisfies

$$\mu(x) = w_Q \pi_Q(x) = \frac{e^{-F_R(x)}}{Z}, \quad \mu(\cdot \mid Q) = \pi_Q, \quad \mu(1) = 1.$$

This choice of weights is a new ensemble hypothesis, not a uniquely derived or physically selected state.

For every allowed root r and $y = rx$, inverse pairing and symmetric mobility give the exact identity

$$\mu(x)c_r(x) = \frac{m_r(x)}{Z} e^{-(F_R(x)+F_R(y))/2} = \mu(y)c_{r^{-1}}(y).$$

Summing against $f(y) - f(x)$ proves $\mu(Lf) = 0$ on the entire finite function domain. The two labelled phase/link channels at $K = 2$ are retained. Gauge transformations preserve the squared covariant differences and the Wilson characters, hence the energy, counting law and finite projection. This does not prove limiting stationarity or an infinite-volume semigroup.

Exact cutoff estimate

Write $z = (j, k, u)$ and let $F_0(z) = F_R(z, 0)$. Erasing occupancy for this comparison is a summation device, not a changed transition or projection. The original energy decomposes as

$$F_R(z, \ell) - F_0(z) = Q \left(\frac{R^4}{4} + \frac{R^6}{6} \right) + \frac{R^2}{2} \sum_v s_v^2 \ell_v + \frac{R^2}{2} \sum_{e=(v,w)} J_e(s) (\ell_w (-1)^{k_w} - U_e \ell_v (-1)^{k_v})^2.$$

Since $s_v \geq 1/2$, binary occupancy and nonnegative squares imply

$$F_R(z, \ell) \geq F_0(z) + Qc(R), \quad c(R) = \frac{R^2}{8} + \frac{R^4}{4} + \frac{R^6}{6}.$$

All aperture and Wilson boundary terms remain in F_0 . Summing over the same z gives $\sum_z e^{-F_R(z, \ell)} \leq e^{-Qc(R)} Z_0$. Let $t = e^{-c(R)} \in (0, 1]$. For any fixed vertex v , $Z \geq Z_0$ gives

$$0 \leq \mu_{n,R}(\ell_v^2) \leq \sum_{\ell: \ell_v=1} t^{\sum \ell} = t(1+t)^{N-1} \leq 2^{N-1} e^{-c(R)}.$$

The binomial sum counts every occupation pattern; phase multiplicities already occur identically in Z_0 . Independently, summing entire charge sectors gives

$$\mu_{n,R}(Q > 0) \leq (1+t)^N - 1 \leq (2^N - 1)t, \quad \ell_v^2 \leq \mathbf{1}_{Q>0}.$$

This provides a second bound with the same conclusion.

For an explicit modulus, $c(R) \geq R/6$ for $R \geq 1$ and $e^c \geq 1 + c$. Therefore

$$\mu_{n,R}(\ell_v^2) \leq \frac{6 \cdot 2^{N-1}}{R}.$$

For any $\delta > 0$, $R \geq \lceil 6 \cdot 2^{N-1} / \delta \rceil + 1$ makes the right side smaller than δ . The modulus depends on n ; no uniform-volume bound or interchange of limits is asserted.

Verdict and verification scope

Both closed $\mathbb{Z}_2$ characters satisfy $H_0^2 = H_1^2 = 1$ pointwise. The ordered squared expectations of the R-488 observables are thus

$$\lim_{n \rightarrow \infty} \lim_{R \rightarrow \infty} (\mu(\ell_a^2), \mu(\ell_d^2), \mu(H_0^2), \mu(H_1^2)) = (0, 0, 1, 1).$$

This rejects the preregistered conjunction requiring all four entries to be positive. It does not exclude a degenerate weak state limit.

Primary verification evaluates the inherited complete energy and root catalog. The independent implementation reconstructs the polynomial coefficients from incidence without importing that implementation. Adversarial controls cover signs, factors, multiplicity and limit order. Lean checks Gibbs algebra, the finite mass bound, exponential estimate, divergence of $c(R)$, the squeeze implication and character squares. The source-to-energy decomposition and full combinatorial argument are given above; finite diagnostic sampling is not their universal proof. The integrated run records exact counts, hashes and compilation output.

Devil’s-advocate review

1. Sign/factor: reversed Gibbs sign and missing sextic or aperture denominators are rejected by exact coefficient controls.
2. Multiplicity: identifying zero-radius phases changes the counting measure. Both lanes retain the coordinate labels and count all grades.
3. Domain: charge erasure is used only for a summation bound. All generator roots still conserve charge and retain their inverse channels.
4. Limit: finite positivity is insufficient. The displayed modulus proves the inner limits; it does not justify reversing the order.
5. Scope: a vanishing radial witness rejects this nondegeneracy target only. The holonomy witnesses remain nonzero. No other ensemble is tested.

These objections are dismissed within the explicitly retained restrictions. External review of multiplicity and limit quantifiers is invited.

Result footer

Result ID: R-508
Claim ID: C6-SPACETIME-SIGNATURE
Precise statement: Ordered squared limits are (0,0,1,1).
Verdict: CANDIDATE_REJECTED
Dependencies: PAH-001; OMC-004/008/010/012/013; OMC-015 prior
Hypotheses: Exactly the preregistered labelled counting domain
Evidence grade: ANALYTIC, EXACT, EXECUTED; Lean cross-check
Scope: fixed OMC-010 path, R before n, counting ensemble
Tier before / after: T1 / T1 (host); PAH-OMC-014 unresolved
No-overclaim statement: No physical or full joint-limit claim
Next required action: Separate owner contract for a new law
Falsification gate: Exact failure of charge-energy/counting bound
Reproduction command: python -X utf8
 verification/scripts/pah_omc015_counting_cutoff_verify.py
Expected output: PASS; CANDIDATE_REJECTED; Lean PASS